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The influence of grant renewal on research content: evidence from NIH-funded PIs

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Abstract

The continuity of research funding is a critical factor influencing scientific progress. Interruptions in funding can disrupt research activities, force researchers to shift their focus, and potentially hinder innovation. Grant renewal, which provides continued funding, is an increasingly important instrument in the research funding landscape. However, its impact on the research content remains underexplored. This study investigates how grant renewal affects the research content of Principal Investigators (PIs) funded by the National Institutes of Health (NIH), specifically examining their effects on three key dimensions of research content: switching probability, diversity, and novelty. To examine the effects of grant renewal on research content, we integrate funding data from NIH ExPORTER, textual data from PubMed and comprehensive publication and author information from SciSciNet. We measure these dimensions of research content using text-based indicators, as text-based methods offer a more nuanced understanding of the essence of research content. Potential selection bias is addressed using Propensity Score Matching (PSM) to create comparable groups of PIs based on grant renewal status following their first R01 grant's initial duration. A Differences-in-Differences (DID) model then estimates the causal impact of grant renewal on research content. Our findings reveal that while grant renewal is associated with a lower probability of PIs switching research content and a reduction in the diversity of their research content, it also has a statistically significant positive effect on research novelty. These results suggest that while grant renewal narrows the thematic scope of PIs' research, it fosters an environment conducive to focused innovation within a chosen area. This study highlights the potential of grant renewal to be an effective instrument for promoting impactful research by enabling PIs to pursue long-term research goals with continued funding support. The findings have significant implications for research funding policy, underscoring the importance of considering the effects of funding mechanisms not just on funding acquisition but also on the content and trajectory of scientific research.

Keywords Grant renewal · Research content · NIH · Text-based · PSM-DID

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Introduction

As a primary input of the basic science, research funding plays a critical role in advancing scientific and technological progress, which in turn has far-reaching implications for economic development, public health, national security, and overall societal well-being (Azoulay et al., 2018; Bush, 1945; Hicks, 2012; Ioannidis, 2011; Jones, 2021; Li et al., 2024; Vasan & West, 2021; Wang & Barabási, 2021). The growing imbalance between the limited availability of research funding and the expanding number of researchers has led to increasingly competitive funding allocation to improve funding efficiency (Abramo et al., 2012; Auranen & Nieminen, 2010; Hicks, 2012; Lewis, 2015; Münch & Baier, 2012; Sandström et al., 2018; Schweiger et al., 2024; Gläser & Whitley, 2007). However, this heightened competition often results in fragmented funding distribution (Aagaard et al., 2020), shorter grant durations, and an elevated risk of funding interruptions (Azoulay et al., 2011; Cocos & Lepori, 2020; Lepori & Reale, 2019; Tham, 2023). Such interruptions have been shown to negatively impact research, leading to increased time spent on grant writing and securing new funding (Dresler et al., 2023; Graves et al., 2011; Herbert et al., 2013; Ioannidis, 2011; Schweiger, 2023; von Hippel & von Hippel, 2015), diminished research activity (Tham, 2023), and a reduction in high-novelty research endeavors (Azoulay et al., 2011; Manso, 2011).

In response to these challenges, funding agencies have employed various funding instruments to enhance funding continuity. One prevalent instrument is the establishment of large grant types, which offer increased funding amounts and extended durations, such as large individual grants or research centre grants (Bloch & Sørensen, 2015; Bloch et al., 2023). For example, the average grant size at the UK Engineering and Physical Sciences Research Council, the Norwegian Research Council, and the Swiss National Science Foundation has increased substantially (Bloch et al., 2023; Ma et al., 2015). Concurrently, Nordic countries, including Denmark, Finland, Norway, and Sweden, have introduced relatively comprehensive Centre of Excellence (CoE) schemes over the past two decades (Aksnes et al., 2012; Bloch et al., 2016; Hellström et al., 2018).

However, large grants also pose risks and burdens to funders, requiring substantial and long-term financial commitments upfront when the likelihood of success is uncertain, along with increased administrative costs (Bloch & Sørensen, 2015; Guthrie et al., 2013). To mitigate these risks, some funding agencies offer a more manageable instrument for ongoing grants of conventional types: grant renewal. For instance, the US National Institutes of Health (NIH) provides renewal opportunities for expiring grants. Similarly, The UK British Heart Foundation employs a five-year rolling funding mechanism for long-term programs (Guthrie et al., 2013). Although less recognized than CoEs or other large grant formats, grant renewal deserves further investigation because it shares more characteristics with conventional grant types, offering funders greater control, flexibility, and cost-effectiveness.

While some research has examined grant renewal, the majority has focused on factors influencing the success of renewal applications. For example, studies by Carter (1974), Maas (2015), and Wang et al. (2022) suggested that grants demonstrating superior performance in the previous funding period are more likely to secure continued funding. Lauer (2016) emphasized the critical role of methodology and significance in renewal application decisions. Greenblatt et al. (2022) found that grants with a higher risk-taking propensity are renewed less frequently than those with a more conservative approach. Unlike previous studies that primarily focus on funding acquisition, Tham (2023) examined the effect of funding renewal timeliness on research activity, finding that the labs experiencing funding interruption had significantly higher spending. This is meaningful because understanding the relationship between research funding instruments and knowledge production is important to assess and optimize instrument effectiveness (Horta & Li, 2023; Schweiger et al., 2024). To delve deeper into the dynamics of knowledge production under continued funding, we focus on research content as a tangible manifestation of researchers' scientific endeavors. Grant renewal is intended to enable researchers to pursue the same long-term goals as the initial grant, albeit with new specific aims.[1] In order to evaluate whether this objective is met, this study focuses on NIH-funded Principal Investigators (PIs) who have received renewed R01 grants. Employing Propensity Score Matching (PSM) and Differences-in-Differences (DID) methods, we analyze the influence of grant renewal on three content-related dimensions reflecting researchers' knowledge production strategies: the switching probability of research content, the diversity of research content, and the novelty of research content (Laudel, 2023).

The remainder of this paper is structured as follows: a literature review analyzing the potential impacts of grant renewal on research content and the measurements of research content characteristics, followed by a description of research sample, data and methods. Subsequent sections present and interpret the empirical results. The final section concludes the paper by summarizing the findings and outlining their policy implications.

Literature review

Potential impacts of grant renewal on research content

Analyzing the potential impacts of grant renewal on research content requires considering how researchers respond to their funding situation. Researchers adapt to changing resource environments by developing a variety of strategies to secure funding and sustain their research, which, in turn, influence the content of their work (Laudel, 2023). As actors with a degree of autonomy and authority over their research activities and funding acquisition (Gläser et al., 2022; Morris, 2003), researchers' primary motivation is to sustain their research by balancing their research and funding portfolios (Gläser, 2019; Gläser et al., 2010; Laudel, 2023; Laudel & Gläser, 2008; Lukkonen & Thomas, 2016).

From the perspective of researchers' responses to their funding situation, the impacts of grant renewal are multifaceted. On the one hand, grant renewal offers a higher success rate of securing continued funding and avoiding funding scarcity. This stable funding environment may reduce the need for researchers to frequently change their research directions to align with funder or reviewer preferences, potentially mitigating the loss of epistemic diversity, risk avoidance, and the tendency to gravitate towards mainstream research (Banal-Estanol et al., 2016; Boudreau et al., 2016; Gläser, 2019; Gläser et al., 2010; Gross & Bergstrom, 2021; Hackett, 1987; Hessels et al., 2011; Laudel, 2006; Leišytė, 2007; Morris, 2000). Furthermore, grant renewal can free researchers from funding searching, allowing them to concentrate on their research. The assurance of adequate funding and research time is a crucial prerequisite for fostering innovation (Azoulay et al., 2011; Heinze et al., 2009; Ioannidis, 2011; Schweiger et al., 2024).

[1] Available at https://www.niaid.nih.gov/grants-contracts/apply-renewal.

On the other hand, grant renewal can also be viewed as a form of thematic funding instrument, as it constrains research topics by reviewing the connection between the prior grant and the proposed one. Traditionally, thematic funding instruments have research themes and objectives defined by funding agencies (Kearnes & Wienroth, 2011; Madsen & Nielsen, 2024; Ramos-Vielba et al., 2022; Sarewitz & Pielke, 2007; Schweiger et al., 2024). Under such thematic funding schemes, researchers tend to change their research interests to align with funder-defined priorities during the funding duration but may revert toward their original focus areas when the funding expires (Gomes, 2019; Madsen & Nielsen, 2024; Myers, 2020). Thematic priorities can also diversify research portfolios by fostering cross-fertilization between researchers' existing topic foci and new topic branches (Laudel, 2023; Luukkonen & Thomas, 2016). However, in the case of grant renewal, the research content before and after renewal is proposed by the investigators themselves. This suggests that the likelihood of change and the diversity of research topics for these researchers may not significantly increase, thus maintaining thematic consistency. This thematic restriction could potentially limit the breadth of knowledge needed for innovation. Existing studies widely regard innovation as a process of knowledge recombination (Bornmann et al., 2019; Fleming, 2001; Foster et al., 2015; Lee et al., 2015; Luo et al., 2022; Tahamtan & Bornmann, 2018; Uzzi et al., 2013; Wang et al., 2017; Zhang et al., 2019). Therefore, researchers funded through renewals might exhibit more conservative knowledge production behaviors to maintain their existing research portfolios, even though the ultimate goal of long-term funding is to foster innovative breakthroughs.

This analysis reveals the multifaceted nature of the impacts of grant renewal on research content. While a stable funding environment provided by grant renewal can empower researchers to pursue consistent research directions, enhance epistemic diversity, and strive for innovation, the thematic constraints inherent in grant renewal may simultaneously limit diversity and novelty. Therefore, empirical investigation is crucial to determine the actual effects of this funding instrument.

Measurements of research content characteristics

Based on the definition of Shi and Evans (2023), content refers to the substance of papers and patents. This study focuses specifically on research content as reflected in scientific publications, including substantive topics, themes, and ideas. Within the framework of scientometrics, we review the methods for measuring research content characteristics.

Measurements of research content characteristics can be broadly categorized as citation-based and text-based. Citation-based approaches typically rely on citation networks or co-citing networks. Citation networks illustrate the relationships between citing and cited papers. For example, Uzzi et al. (2013) defined novel papers as those citing an unusual combination of references, which rarely appear together in other papers. Matsumoto et al. (2021) visualized citation relationships to calculate a novelty score based on citation similarity. Funk and Owen-Smith (2017) developed the disruption index to measure the extent to which a new invention consolidates or destabilizes existing technology streams; Wu et al. (2019) further applied this index in the scientific domain, defining disruptive papers as those that receive more citations than their references. Co-citing networks, on the other hand, represent all publications of a specific entity as nodes, with the frequency of co-citation between each pair of publications serving as the edge weight. Community detection algorithms can then be applied to these networks to identify potential research topics (Zeng et al., 2019). For instance, Zeng et al. (2019) used this approach to calculate the probability of scientists switching between research topics throughout their careers. Guo et al. (2024) employed this method to calculate scientists' topic entropy, a measure of research focus diversity.

Text-based measurements initially relied on keyword analysis but have advanced significantly with the development of Natural Language Processing (NLP) techniques. The emergence of keyword systems within community-curated ontologies, such as Medical Subject Heading (MeSH) terms for MEDLINE papers and Physics and Astronomy Classification Scheme (PACS) codes for APS papers, has greatly enhanced the feasibility of content analysis (Shi & Evans, 2023). For example, Azoulay et al. (2011) measured journal article novelty based on the average age of MeSH terms, Boudreau et al. (2016) assessed research grant proposal novelty using new combinations of MeSH terms, and Jia et al. (2017) quantified patterns of research interest evolution using topic vector similarity based on PACS codes. More recently, NLP methods operating at various level, including Word2vec for word-level semantic vector representations (Mikolov et al., 2013), SciBERT for sentence-level representations (Beltagy et al., 2019), and SPECTER for document-level representations (Cohan et al., 2020), among others, have been employed to measure research content characteristics. For instance, Yu et al. (2023) analyzed scholars' research diversity through multi-dimensional calculation of knowledge entities. Wang et al. (2023a) introduced a knowledge entity-based disruption indicator that quantifies the impact of scientific breakthroughs on knowledge evolution. Yang et al. (2024) developed a deep learning-based method to predict the emergence of research topics. Wang et al. (2023b) measured topic interdisciplinarity using the BERTopic model. Several studies have also explored scientific novelty using various semantic embedding techniques (Hou et al., 2022; Luo et al., 2022; Shibayama et al., 2021; Yan & Fan, 2024). Compared to citation-based approaches, text-based methods offer a more nuanced understanding of the essence of research content. Therefore, this study will utilize text-based methods to measure the switching probability, diversity, and novelty of research content among NIH-funded PIs.

Data and methods

Overview of NIH grant renewal instrument

NIH regulations allow researchers to apply for continued funding through a renewal application as their existing grant period nears its end.[2] This option is available for main grant types, such as the R01. NIH extramural funding encompasses two primary application types: new and renewal. New applications are intended for researchers proposing entirely new research goals, while renewal applications are designed for investigators pursuing new specific aims within a broader, long-term research goal.

Both application types generally follow the same NIH application process, review framework and pay lines, with two main exceptions. First, in the application stage, in addition to standard application materials, renewal applications require a separate Research Performance Progress Report (RPPR). The RPPR details the original grant's goals and the

[2] Available at https://grants.nih.gov/grants/how-to-apply-application-guide/prepare-to-apply-and-register/type-of-applications.htm.

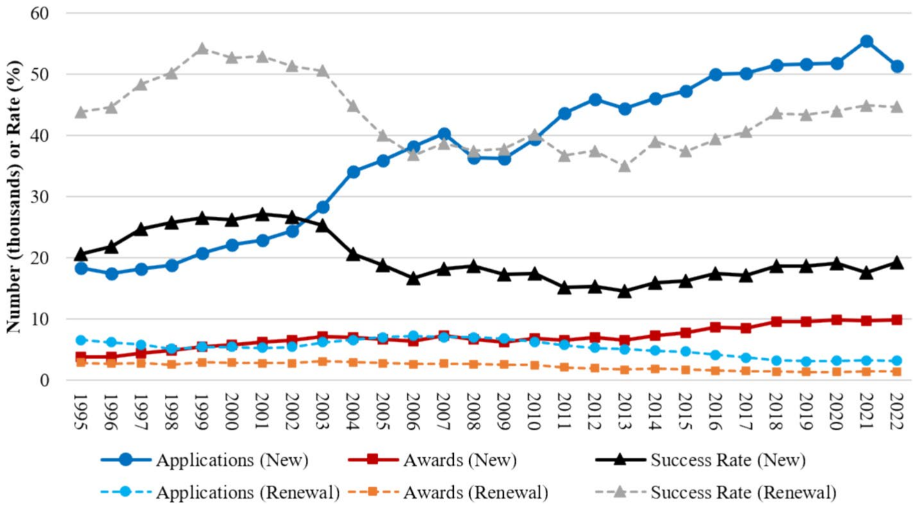

Fig. 1 Comparisons of new and renewal application types: applications, awards, and success rates, 1995–2022. *Sources* https://report.nih.gov/reportweb/web/displayreport?rid=565, https://nexus.od.nih.gov/all/2016/05/26/grant-renewal-success-rates-then-and-now/

progress achieved toward those goals.[3] Second, during the review stage, an additional criterion of "renewal or not" is considered. While not scored separately, this criterion influences the overall score based on the proposed research's continuity with the previously funded grant.[4]

Despite the similarities in the funding process, renewal applications have a significantly higher success rate than new applications. Research Project Grants (RPGs), the most common type of NIH extramural funding by both quantity and total funds,[5] illustrate this disparity. Between 1995 and 2022, the NIH received an annual average of 37,175 new RPG applications, awarding 7027 annually, for a success rate of 19.91%. In contrast, the annual average number of renewal RPG applications was 5345, with 2301 awarded annually, resulting in a success rate of 43.24%. Of the 161,187 RPGs funded during this period, renewals comprised 32.75% of the annual average, yet their success rate was 2.17 times higher than that of new grants. This disparity highlights a clear NIH funding emphasis on grant renewals (Fig. 1).References Whitley and Gläser (2007), Gläser et al. (2010), Yan et al. (2024) were mentioned in the manuscript; however, these were not included in the reference list. As a rule, all mentioned references should be present in the reference list. Please provide the reference details to be inserted in the reference list. Thank you for pointing out the issues with the references. We have addressed them as follows:The citation Whitley & Gläser (2007) was an error in the author order. It should be Gläser & Whitley

[3] Available at https://grants.nih.gov/grants/rppr/index.htm and https://www.niaid.nih.gov/grants-contracts/apply-renewal#A6.

[4] Available at https://grants.nih.gov/policy/peer/simplifying-review/framework.htm.

[5] The data show that in 2022, the number of projects awarded by RPGs accounted for 79% of the total number of all NIH Research Grants awarded in the same year (https://report.nih.gov/nihdatabook/report/23), and the total funding for RPGs comprised 77% of the total funding of all NIH Research Grants awarded in the same year (https://report.nih.gov/nihdatabook/report/24).

(2007), which was already included in the reference list. We have corrected the in-text citation accordingly.The reference Gläser et al. (2010) was included in the reference list, but contained a typo (Glaser instead of Gläser). We have corrected the spelling in the list.The reference Yan et al. (2024) was also correctly included in the list. The issue was the format of the in-text citation; we have corrected it from Yan et al. (2024) to Yan & Fan (2024).

Data collection and sample construction

This study utilized three primary data sources: the NIH ExPORTER database, which provides comprehensive information on NIH-funded grants since 1985[6]; the SciSciNet database, a large-scale, open-access resource for science of science research containing over 134 million scientific publications and extensive linkages to funding and public uses (Lin et al., 2023)[7]; and PubMed, a comprehensive bibliographic database of biomedical literature containing over 37 million citations.[8]

Using the NIH ExPORTER database, we first extracted linkages between each NIH-funded PI ID and their corresponding Grant IDs, which are split into annual records by NIH. We focused on 67,138 PIs who had received at least one new R01 grant with the duration ≥ 3 years. To investigate the potential influence of grant renewal, we defined the **treatment group** as PIs who received at least a 3-year renewal immediately following the initial 3–5 year duration of their **first** career R01 grant, without any funding interruption.[9] The **control group** was defined as PIs who experienced at least a 3-year gap in NIH funding after the end of their **first** R01 grant's initial duration. Furthermore, we restricted both groups to PIs who had not received any NIH grants other than their first R01 grant during its initial duration and, for the treatment group, the subsequent renewal. This resulted in a preliminary sample of 1756 PIs in the treatment group and 14,228 PIs in the control group.

We then utilized the SciSciNet database to match SciSciNet Author IDs with NIH PI IDs based on PI names. SciSciNet provides linkages between funded projects and papers (SciSciNet Paper ID to NIH Grant ID), papers and authors (SciSciNet Paper ID to SciSciNet Author ID), and author IDs and names (SciSciNet Author ID to SciSciNet Author Name). For each PI in our preliminary sample, we calculated the name similarity between their NIH ExPORTER record and all author names associated with papers funded by that PI's grants in SciSciNet. The pair with the highest similarity score was selected as the match. This process was performed using the TheFuzz package.[10] PIs without SciSciNet records or with a maximum similarity score < 80 (out of 100) were excluded. This refined our sample to 1678 PIs in the treatment group and 11,385 PIs in the control group.

To calculate covariates and dependent variables, and ensure that PIs in the control group remained active in research when funding interrupts, we further required all PIs to have ≥ 2 PubMed-indexed publications in each of the following periods: the 3 years before the start of the first R01 grant's initial duration, during the initial duration of the first R01 grant,

[6] Available at https://reporter.nih.gov/exporter/projects.

[7] Available at https://springernature.figshare.com/collections/SciSciNet_A_large-scale_open_data_lake_for_the_science_of_science_research/6076908/1.

[8] Available at https://pubmed.ncbi.nlm.nih.gov/.

[9] The reason we focus on the renewal status at the end of the initial duration of the PIs' first R01 grant is to minimize potential confounding effects from other funding. This also represents the point at which the influence of grant renewal on research content is likely most significant.

[10] Available at https://github.com/seatgeek/thefuzz.

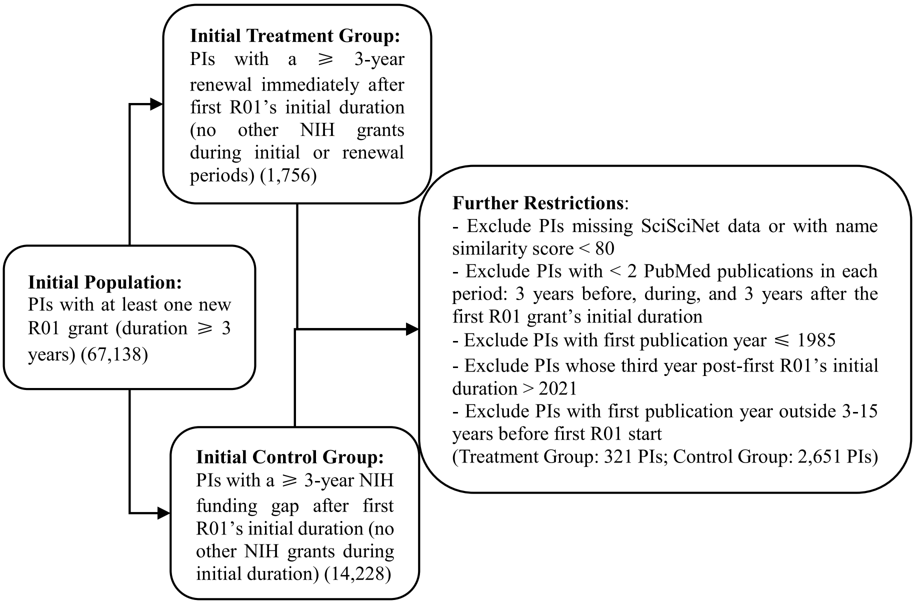

Fig. 2 The process of sample construction

and the 3 years after the end of the first R01 grant's initial duration. Additionally, to avoid including PIs with NIH funding records before 1985 or PIs with incomplete observations, we restricted the sample to PIs whose first publication year was > 1985 and whose third year after the end of their first R01 grant's initial duration was ≤ 2021.[11] To control for potential confounding effects, we also limited the sample to PIs whose first publication year was between 3 and 15 years before the start of their first R01 grant. These criteria resulted in a sample of 321 PIs in the treatment group and 2,651 PIs in the control group, which was used for subsequent 1:1 matching of treatment and control group members.

Finally, we used the Bio.Entrez package[12] to retrieve the textual information for each publication in the sample form the PubMed database. This information, including MeSH terms, titles and abstracts, will be used to calculate the dependent variables. Figure 2 illustrates the complete sample construction process.

Measurements of dependent variables

Switching probability of research content

To quantify the switching probability of research content, we employed the SPECTER model to generate document-level embedding vector representations for each publication in our sample, based on the publication's title and abstract. SPECTER leverages

[11] This is because the SciSciNet database is based on the Microsoft Academic Graph (MAG) dataset, which was last updated on December 6, 2021.

[12] Available at https://biopython.org/docs/1.75/api/Bio.Entrez.html.

inter-document context within a Transformer-based architecture to produce rich representations suitable for various downstream tasks without requiring fine-tuning (Cohan et al., 2020).

Using these embeddings, we calculated the similarity between the sets of publications from different periods for each PI: specifically, between the 3 years before the first R01 grant and its initial duration, and between the first R01 grant's initial duration and the 3 years following its end. These calculations were performed using the Average Maximum Match (AMM) method, defined as follows:

$$Sim\left(V,V'\right)=\frac{\sum_{v\in V}\max_{v'\in V'}Sim\left(v,v'\right)+\sum_{v'\in V'}\max_{v\in V}Sim\left(v,v'\right)}{|V|+|V'|}$$

In this formula, V and V' represent the sets of semantic vectors at two different publication periods. $|V|$ and $|V'|$ denote the cardinalities of these sets (i.e., the number of vectors in each set). v and v' are individual vectors within sets V and V', respectively. $Sim\left(v,v'\right)$ is the cosine similarity between two vectors, calculated as:

$$Sim\left(v,v'\right)=\frac{\sum_{k=1}^{l}\left(v(k)\times v'(k)\right)}{\sqrt{\sum_{k=1}^{l}v(k)^2}\times\sqrt{\sum_{k=1}^{l}v'(k)^2}}$$

where k denotes the k-th element of the vector, and l represents the dimensionality of the vector.

Finally, following Jia et al. (2017), the *switching probability of research content* between two periods is defined as $1-Sim\left(V,V'\right)$.

Diversity of research content

We define the *diversity* of a PI's research content within a specific publication period as the average dissimilarity between all possible pairs of publications within that period. This dissimilarity was calculated as the cosine distance (1 minus the cosine similarity) between the SPECTER-generated vector representations of each publication pair.

Novelty of research content

We measured the *novelty of research content* following the methods proposed by Azoulay et al. (2011) and Boudreau et al. (2016). Azoulay et al. (2011) measured publication novelty based on the average age of single MeSH terms (the number of years since its first appearance), while Boudreau et al. (2016) assessed proposal novelty by considering the frequency of previously unseen MeSH term pairs in the scientific literature. We adopted the common notion that novelty arises from unusual combinations of knowledge. Therefore, we quantified the novelty of each publication based on the average age of MeSH term pairs, and converting a negative indicator to a positive one (Yang et al., 2022). The indicator is defined as follows:

$$Novelty=-1\times\ln\left(\frac{\sum Age}{n(n-1)/2}+1\right)$$

where *Age* represents the age of a specific MeSH term pair, calculated as the number of years since its first appearance in PubMed database, based on the information from the MEDLINE Co-Occurrences (MRCOC) Files.[13] n represents the counts of single MeSH term of the publication.

We then measured the *novelty* of a PI's research content within a specific publication period by averaging the novelty scores of all publications of that period.

Propensity score matching

The previously constructed research sample reflects two potential effects: a selection effect, where the grant renewal instrument tends to select PIs with distinct knowledge production strategies, and a treatment effect, where the grant renewal instrument directly influences research content. To isolate the net treatment effect, we employed PSM method to mitigate potential endogeneity in the model. PSM is a statistical matching technique that estimates the effect of an intervention by accounting for the covariates that predict receiving the treatment. This method helps reduce bias from confounding variables that might otherwise skew treatment effect estimates obtained by simply comparing outcomes between treated and untreated groups (Rosenbaum & Rubin, 1983).

In this study, we selected the following covariates: ***career start year***, defined as PI's first publication year; ***prior NIH funding***, defined as the natural logarithm of the total NIH funding received before and during PI's first R01 grant; ***time from career start to first R01 grant***, defined as the number of years between the PI's first publication year and the start of his first R01 grant; ***duration of the first R01 grant*** (3, 4, or 5 years), represented by two dummy variables; ***prior number of publications***, defined as the natural logarithm of total number of publications published before and during the initial duration of PI's first R01 grant; ***prior average citation count***, defined as the natural logarithm of average citation count of publications published before and during the initial duration of PI's first R01 grant; ***prior average reference count***, defined as the natural logarithm of average reference count of publications published before and during the initial duration of PI's first R01 grant; ***prior average team size***, defined as the natural logarithm of average author count of publications published before and during the initial duration of PI's first R01 grant; ***prior content switching probability***, defined as the switching probability of research content between the three years before PI's first R01 grant and its initial duration; ***prior content diversity***, defined as the diversity of research content during the initial duration of PI's first R01 grant; and ***prior content novelty***, defined as the novelty of research content during the initial duration of PI's first R01 grant.

After defining these covariates, we applied one-to-one nearest neighbor propensity score matching without replacement, restricted to the common support, to match each PI in the treatment group with the most similar PI from the control group, thus forming the final matched control group for subsequent analysis. On the basis of PSM, although we could not completely rule out a selection effect, the differences were more likely attributable to the treatment effects after matching.

[13] Available at https://lhncbc.nlm.nih.gov/ii/information/MRCOC.html.

Table 1 Logistic regression results before PSM

Variables	Treat
Career start year	−0.0839*** (0.0184)
Prior NIH funding	−0.102 (0.135)
Time from career start to first R01 grant	−0.0843*** (0.0236)
Duration of the first R01 grant	
Dummy variable 1 (0: 3 years; 1: 4 years)	0.486** (0.191)
Dummy variable 2 (0: 3 years; 1: 5 years)	0.957*** (0.187)
Prior number of publications	0.621*** (0.139)
Prior average citation count	0.349*** (0.101)
Prior average reference count	0.359* (0.189)
Prior average team size	−0.992*** (0.220)
Prior content switching probability	−1.891 (1.998)
Prior content diversity	−5.533*** (1.359)
Prior content novelty	1.723*** (0.304)
Constant	170.557*** (35.497)
N	2948
Pseudo R^2	0.165

Robust standard errors are reported in parentheses

***$p<.01$, **$p<.05$, *$p<.1$

Differences-in-differences

To estimate the net effect of grant renewal on research content, we employed a DID model with two-way fixed effects (TWFE). The impact of grant renewal was estimated using the following regression equation:

$$Y_{it} = \alpha + \beta Treat_i \times Post_t + PI_i + Period_t + \delta X_{it} + \varepsilon_{it}$$

where Y_{it} represents the dependent variable for PI i in time period t: either the *switching probability*, *diversity*, or *novelty of research content*. $Treat_i$ is a dummy variable that equals one if PI i received at least a 3-year renewal immediately following the initial duration of its first career R01 grant and zero otherwise. $Post_t$ is a dummy variable that equals one if time period t after the treatment. To ensure reliable measurement of research content characteristics, we aggregated the observation years into two time periods: the period during the initial duration of PI's first R01 grant, and the period encompassing the three years following the end of the initial duration. PI fixed effects PI_i capture time-invariant heterogeneity across PIs. Time fixed effects $Period_t$ capture individual-invariant heterogeneity across time periods. This TWFE specification is mathematically equivalent to a model including separate *Treat* and *Post* main effects and provides a more robust estimate of the causal effect by controlling for unobserved heterogeneity (Wing et al., 2018). The coefficient of interest, β, represents the estimated DID effect of grant renewal on research content. X_{it} represents a vector of four control variables for PI i in time period t: *number of publications*, *average citation count*, *average reference count*, and *average team size* (all transformed using the natural logarithm).

Results

Propensity score matching results

Tables 1 presents the logistic regression results before PSM, where the dependent variable is *Treat* and the independent variables are the covariates described previously. With the exception of the regression coefficients for *prior NIH funding* and *prior content switching probability*, all other covariates show significant differences between the treatment and control groups, suggesting potential selection bias prior to grant renewal. Specifically, the treatment group exhibit a lower *career start year*, shorter *time from career start to first R01 grant*, smaller *prior average team size*, and lower *prior content diversity* compared to the control group. Conversely, the treatment group has higher *prior number of publications*, *prior average citation count*, *prior average reference count*, and *prior content novelty* than the control group.

Table 2 displays the covariate balance testing results after 1:1 PSM. After matching, the absolute standardized mean difference (|%bias|) for nearly all covariates is ≤ 10% (with the

Table 2 Covariate balance testing results after 1:1 PSM

Variables	Mean		%bias	*t*-test	
	Treated	Control		t	$p>\|t\|$
Career start year	1992.4	1992.7	−4.4	−0.58	0.565
Prior NIH funding	13.844	13.816	5.0	0.62	0.533
Time from career start to first R01 grant	8.680	8.348	10.2	1.32	0.186
Duration of the first R01 grant					
Dummy variable 1 (0: 3 years; 1: 4 years)	0.295	0.298	−0.7	−0.09	0.931
Dummy variable 2 (0: 3 years; 1: 5 years)	0.505	0.505	0.0	−0.00	1.000
Prior number of publications	3.155	3.137	3.0	0.38	0.702
Prior average citation count	4.517	4.457	8.4	1.02	0.307
Prior average reference count	3.521	3.486	8.0	1.02	0.310
Prior average team size	1.492	1.488	1.0	0.14	0.892
Prior content switching probability	0.144	0.144	0.8	0.10	0.917
Prior content diversity	0.210	0.212	−3.1	−0.39	0.696
Prior content novelty	−3.009	−3.01	0.3	0.03	0.976

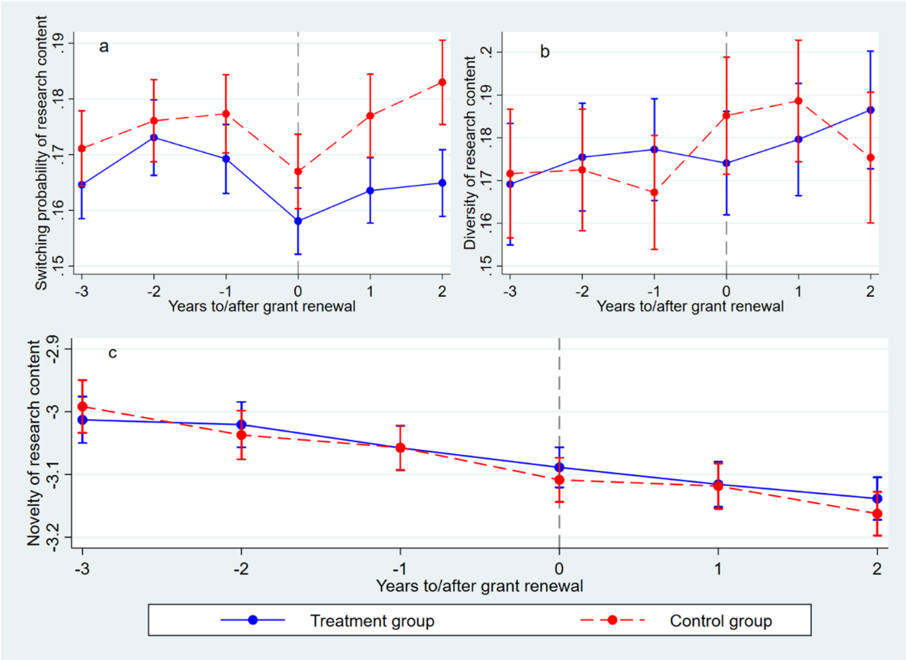

Fig. 3 The results of parallel trends analysis of dependent variables. *Note* Error bars represent 95% confidence intervals

exception of *time from career start to first R01 grant* at 10.2%). Furthermore, all t-tests for covariate balance are statistically insignificant, indicating successful balance between the treatment and control groups after matching. The average absolute standardized mean difference across all covariates is 3.4% (< 5%), and the overall p-value for the balance test is 0.985 (> 0.1), further supporting the effectiveness of the matching procedure.

While the PSM procedure effectively addresses observable differences between the treatment and control groups, a key assumption of the DID approach is that, in the absence of treatment, both groups would have followed parallel trends in the outcome variables. To visually assess the plausibility of this assumption, we conducted a parallel trends analysis for each of our three dependent variables: switching probability of research content, diversity of research content, and novelty of research content. Figure 3 presents the results of this analysis, plotting the average values of each dependent variable for both the treatment and control groups across the years before and after the start year of the potential grant renewal duration (year '0' on the x-axis).

For switching probability (Fig. 3a), both groups exhibit similar trends before the start year of the potential grant renewal duration. However, in the years following this point, the control group shows a more pronounced increase in switching probability. For diversity of research content (Fig. 3b), the values are relatively similar before the start year of the potential grant renewal duration, with the control group even exhibiting slightly lower levels. Subsequently, the control group shows a clear increasing trend in diversity compared to the treatment group, although there's a decrease in the third year of the potential grant renewal duration, indicating that PIs untreated build stable research themes in new research

Table 3 Descriptive statistics after PSM

	N		Mean		Std. Dev		Min		Max	
	Treatment group	Control group	Treatment group	Control group	Treatment group	Control group	Treatment group	Control group	Treatment group	Control group
Dependent variables										
Switching probability of research content	638	638	0.139	0.145	0.040	0.042	0.054	0.047	0.377	0.423
Diversity of research content	638	638	0.214	0.224	0.059	0.067	0.058	0.031	0.387	0.451
Novelty of research content	638	637	−3.058	−3.070	0.282	0.287	−3.795	−3.772	−2.266	−2.224
Control variables										
Number of publications	638	638	2.235	2.100	0.656	0.768	0.693	0.693	4.078	4.754
Average citation count	638	638	4.160	3.992	0.847	0.864	1.253	0.965	9.223	7.621
Average reference count	637	638	3.688	3.616	0.476	0.477	0.288	0.981	5.038	5.409
Average team size	638	638	1.547	1.555	0.416	0.429	0.560	0	3.710	3.045

Table 4 Regression results from the DID model

Variables	(1) Switching probability of research content	(2) Diversity of research content	(3) Novelty of research content
Treat × *Post* = (*TreatRenewal*)	−0.0111***	−0.0136***	0.0238**
	(0.00312)	(0.00492)	(0.0110)
Number of publications	−0.00713***	0.00537	0.00357
	(0.00262)	(0.00450)	(0.0103)
Average citation count	0.00411	0.000510	0.00147
	(0.00290)	(0.00403)	(0.00807)
Average reference count	−0.00708	−0.0227***	0.00183
	(0.00477)	(0.00874)	(0.0145)
Average team size	0.0119**	0.0233***	0.0291
	(0.00512)	(0.00895)	(0.0200)
PI FE	YES	YES	YES
Time FE	YES	YES	YES
Constant	0.151***	0.255***	−3.135***
	(0.0224)	(0.0382)	(0.0671)
N	1274	1274	1272
R^2	0.777	0.763	0.941
Adjusted R^2	0.550	0.522	0.880
Adjusted within R^2	0.0455	0.0531	0.00435

Robust standard errors, clustered at the PI level, are reported in parentheses

***$p<.01$, **$p<.05$, *$p<.1$

areas. For novelty of research content (Fig. 3c), both groups display similar decreasing trend and novelty scores before and during the potential grant renewal duration. It is important to note that, while the raw trends for novelty appear visually similar, our subsequent DID analysis (presented in Table 4) reveals that, after controlling for relevant covariates, the treatment group exhibits statistically significantly higher novelty scores post-treatment compared to the control group.

Differences-in-differences results

Table 3 presents the descriptive statistics of the dependent and control variables after PSM. Regarding the dependent variables, the treatment group has a lower mean *switching probability* and *diversity of research content* compared to the control group. However, the treatment group exhibit a higher mean *novelty of research content*. For the control variables, the treatment group has a higher mean *number of publications*, *average citation count*, and *average reference count*, but a lower mean *average team size* compared to the control group.

Table 4 displays the regression results from the DID model. The coefficients for *TreatRenewal* in column (1) and (2) are negative and statistically significant ($p \leq .01$),

Table 5 Results of robustness checks

Variables	(1) Switching probability of research content	(2) Diversity of research content	(3) Novelty of research content	(4) Switching probability of research content	(5) Diversity of research content	(6) Novelty of research content
Treat × Post = (TreatRenewal)	−0.00953***	−0.0118**	0.0210*	−0.0102***	−0.0117**	0.0303***
	(0.00317)	(0.00500)	(0.0120)	(0.00321)	(0.00512)	(0.0116)
Number of publications	−0.00587*	0.0129**	0.0136	−0.00500**	0.00977**	−0.00758
	(0.00299)	(0.00516)	(0.0101)	(0.00248)	(0.00463)	(0.00974)
Average citation count	0.00249	−0.00316	0.0154	0.000690	−0.00425	0.00511
	(0.00288)	(0.00444)	(0.0106)	(0.00254)	(0.00416)	(0.00914)
Average reference count	−0.00387	−0.0242***	−0.0178	−0.00658	−0.0160**	−0.00194
	(0.00474)	(0.00860)	(0.0164)	(0.00473)	(0.00796)	(0.0144)
Average team size	0.00401	0.0185**	0.0355	0.00267	0.0160**	0.0326
	(0.00480)	(0.00717)	(0.0219)	(0.00475)	(0.00746)	(0.0202)
PI FE	YES	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES	YES
Constant	0.151***	0.262***	−3.191***	0.173***	0.251***	−3.123***
	(0.0214)	(0.0369)	(0.0723)	(0.0221)	(0.0337)	(0.0621)
N	1010	1010	1010	1206	1206	1206
R^2	0.777	0.796	0.941	0.777	0.756	0.937
Adjusted R^2	0.548	0.587	0.881	0.551	0.508	0.873
Adjusted within R^2	0.0202	0.0651	0.0210	0.0207	0.0368	0.0110

Robust standard errors, clustered at the PI level, are reported in parentheses

***$p < .01$, **$p < .05$, *$p < .1$

indicating that grant renewal has a significant negative effect on *switching probability* and *diversity of research content*. Conversely, the coefficient for *TreatRenewal* in column (3) is positive and statistically significant ($p = .031$), suggesting that grant renewal has a significant positive effect on *novelty of research content*. These findings suggest that while the grant renewal instrument restricts the thematic breadth of PIs' research by encouraging them to focus on a specific area, it does not hinder, and in fact appears to promote, innovation within that chosen area. This aligns with the intended purpose of grant renewal, which aims to provide sustained funding for PIs to delve deeper into a selected topic. By reducing the need to constantly seek new funding avenues and shift research directions, grant renewals allow PIs to concentrate their efforts, build upon existing knowledge, and make more substantial progress toward their long-term research goals. This sustained focus, coupled with the potential for deeper exploration within a specific domain, could foster the development of novel ideas and research contributions, as reflected in the positive effect on research content novelty.

Robustness checks

To assess the robustness of our findings, we conducted two additional analyses with restricted samples. First, we excluded PIs whose first R01 grant had an initial duration of 3 years, as this could be considered a relatively short duration compared to the typical 4 or 5-year duration. Second, we excluded PIs whose sub-funding agency within NIH had fewer than 50 PIs in the sample to reduce potential heterogeneity across disciplines. For each restricted sample, we re-ran the PSM and DID analyses. Table 5 presents the results of these robustness checks. Columns (1)–(3) display the DID results for the first robustness check (excluding 3-year grants), and columns (4)–(6) display the results for the second robustness check (excluding small sub-funding agencies). The results are consistent with those of the main model, with statistically significant coefficients in the same directions, thus bolstering the robustness of our findings.

Conclusion

The stability and continuity of research funding are critical factors influencing the productivity and impact of scientific research. Interruptions in funding can disrupt research activities, force researchers to shift their focus, and potentially hinder innovation. Grant renewal, as a mechanism for providing sustained funding, has become an increasingly important instrument in the research funding landscape. This study aims to understand the effects of such funding continuity, provided by grant renewal, on the content of research. Using a combination of PSM and DID analysis, we examine how NIH grant renewal influences three key dimensions of PIs' research content: switching probability, diversity, and novelty.

Our findings reveal a nuanced relationship between grant renewal and research content dynamics. We find that while grant renewal is associated with a reduced likelihood of PIs switching research content and a decrease in the diversity of their research content, it also has a significant positive effect on research novelty. This suggests that the grant renewal instrument, while potentially narrowing the thematic breadth of a PI's research, fosters an environment conducive to focused innovation within a chosen area. These results align with the intended purpose of grant renewal, which is to provide sustained support for PIs to pursue long-term research goals. By alleviating the pressure of constantly seeking new funding and potentially shifting research directions, grant renewal appears to enable PIs to concentrate their efforts, build upon existing knowledge, and make more substantial progress, ultimately leading to increased novelty in their research output. This is consistent with the work of Azoulay et al. (2011) who showed the importance of stable and sustained funding for fostering scientific novelty.

This study makes several contributions to the literature on research funding and its impact on scientific progress. First, it provides empirical evidence on the effects of a specific funding instrument—grant renewal—on the content of research, an area that has been relatively understudied compared to the factors influencing funding allocation (Arora-Jonsson et al., 2023; Kivistö & Mathies, 2023; Sivertsen, 2023). Second, it employs a robust methodological approach, combining PSM and DID analysis to address potential selection bias and isolate the net treatment effect of grant renewal. This allows for a more accurate estimation of the causal impact of this funding instrument. Third, by examining multiple dimensions of research content, including switching probability, diversity, and novelty, our study offers a more comprehensive understanding of the complex interplay between funding instruments and research dynamics. Fourth, unlike previous studies that primarily focus on funding acquisition (Carter, 1974; Greenblatt et al., 2022; Lauer, 2016; Maas, 2015; Wang et al., 2022), this study investigates the impact of grant renewal on research content, providing a new perspective for optimizing funding instruments.

Despite these contributions, this study has limitations. Our analysis primarily focuses on the relatively short-term effects of grant renewal, examining research content dynamics within a 3-year window following the initial grant period. Whether the observed increase in novelty translates into long-term scientific breakthroughs remains an open question. Future research should investigate the longer-term impacts of grant renewal on research productivity, impact, and the advancement of scientific knowledge. As pointed out by research from Myers (2020) and Wang and Barabási (2021), the accumulation of scientific knowledge is a long-term and complex process, where short-term indicators may not always align with ultimate scientific contributions.

The findings of this study have important implications for research funding policy. Our results suggest that grant renewal can be an effective instrument for fostering innovation by enabling focused and continuous research efforts. Funding agencies should consider the potential benefits of grant renewal in promoting not only research continuity but also the generation of novel research outcomes. While concerns about potentially narrowing the scope of research are valid, our findings indicate that the positive effects on novelty may outweigh these concerns. This study lends support to the importance of funding mechanisms that provide sustained support for researchers, in line with the arguments made by advocates for long-term funding instruments such as the "Centre of Excellence" schemes adopted in Nordic countries (Aksnes et al., 2012; Bloch et al., 2016; Hellström et al., 2018). However, it is crucial to strike a balance between providing continued funding and ensuring that funding portfolios remain diverse and support a wide range of research areas. Funding agencies might consider implementing mechanisms to monitor the long-term impacts of grant renewals and ensure that they do not inadvertently stifle exploration of new research avenues. Regular evaluation of funding portfolios and their effects on the research landscape will be essential to optimize the effectiveness of funding instruments and promote a healthy and dynamic scientific ecosystem.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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